

4.1 Overview

Lecture 3 introduced the stochastic multi-armed bandit problem and regret, and ended with a concrete question:

How many times must we sample a suboptimal arm before we know enough to stop?

Lectures 4–5 answer this question in two stages:

Confidence and exploration.

Quantify statistical uncertainty, use it to analyze a simple explore-then-commit policy, and see how confidence intervals can guide exploration.

UCB and regret.

Make all the confidence statements needed by an adaptive algorithm valid at once, then turn optimism into a formal policy and a regret guarantee.

The economic question remains the one from Lecture 3: how should a seller learn about demand while its prices are simultaneously earning revenue? The first step is statistical. We need to understand what an empirical mean does—and does not—tell us after finitely many observations.

The unit follows the progression

CLT scale $\rightarrow$ Hoeffding $\rightarrow$ explore then commit $\rightarrow$ confidence intervals
 $\rightarrow$ simultaneous clean event $\rightarrow$ UCB $\rightarrow \Delta_i \leq 2r_i \rightarrow$ regret and pricing.

Sections marked * are optional reading.

4.2 From CLT intuition to finite-sample confidence

Consider iid random variables $X_1, \dots, X_n$ with mean μ and variance σ^2 . The central limit theorem suggests the informal approximation

$$\sum_{s=1}^n X_s \approx n\mu + \sqrt{n} \sigma Z, \quad Z \sim \mathcal{N}(0, 1).$$

Dividing by n , the empirical mean

$$\widehat{M}_n = \frac{1}{n} \sum_{s=1}^n X_s$$

behaves approximately like

$$\widehat{M}_n \approx \mu + \frac{\sigma}{\sqrt{n}} Z. \quad (4.1)$$

For a Bernoulli random variable with mean μ ,

$$\sigma^2 = \mu(1 - \mu) \leq \frac{1}{4}.$$

The important message is the scale: typical estimation error is of order $1/\sqrt{n}$.

The CLT is an approximation, however, and sequential decisions require a more operational statement:

With probability at least $1 - \delta$, how far can the empirical mean be from the truth after a finite number of samples?

4.2.1 Hoeffding's inequality

Suppose now that $X_1, \dots, X_n$ are iid, take values in $[0, 1]$, and have mean μ . Hoeffding's inequality states that

$$\mathbb{P} \left(|\widehat{M}_n - \mu| \geq r \right) \leq 2e^{-2nr^2}. \quad (4.2)$$

Setting the right-hand side equal to δ gives

$$r(n, \delta) = \sqrt{\frac{\log(2/\delta)}{2n}}, \quad (4.3)$$

and therefore

$$\mathbb{P} \left(|\widehat{M}_n - \mu| \leq \sqrt{\frac{\log(2/\delta)}{2n}} \right) \geq 1 - \delta. \quad (4.4)$$

The CLT and Hoeffding share the same $1/\sqrt{n}$ scale, but answer different questions:

CLT intuition	Hoeffding guarantee
approximate	rigorous upper bound
asymptotic	finite sample
variance-sensitive	valid for any reward in $[0, 1]$
error scale $1/\sqrt{n}$	error scale $\sqrt{\log(1/\delta)/n}$

Four times as many observations are needed to cut uncertainty roughly in half. Demanding a smaller failure probability costs only a logarithmic factor.

4.3 A first algorithm: explore then commit

Recall the stochastic bandit setting: there are K arms and T rounds, and arm i has an unknown mean reward μ_i . Hoeffding already suggests a simple policy when T is known. Begin with two arms satisfying

$$\mu_1 > \mu_2, \quad \Delta = \mu_1 - \mu_2 > 0.$$

Explore then commit:

Sample each arm m times, compute its empirical mean, and then play the arm with the larger empirical mean for every remaining round.

Let $\widehat{M}_{1,m}$ and $\widehat{M}_{2,m}$ denote the two empirical means. If

$$|\widehat{M}_{1,m} - \mu_1| < \frac{\Delta}{2} \quad \text{and} \quad |\widehat{M}_{2,m} - \mu_2| < \frac{\Delta}{2},$$

then $\widehat{M}_{1,m} > \widehat{M}_{2,m}$, so the final commitment is correct. For each arm, Hoeffding gives

$$\mathbb{P} \left(|\widehat{M}_{i,m} - \mu_i| \geq \frac{\Delta}{2} \right) \leq 2 \exp \left(-\frac{m\Delta^2}{2} \right). \quad (4.5)$$

4.3.1 The union bound

For any events $A_1, \dots, A_J$,

$$\mathbb{P} \left(\bigcup_{j=1}^J A_j \right) \leq \sum_{j=1}^J \mathbb{P}(A_j). \quad (4.6)$$

No independence is required. If many confidence statements could fail, make each individual failure probability small enough that their total is still small.

In explore then commit, a wrong commitment requires at least one of the two deviation events in (4.5). Hence

$$\mathbb{P}(\text{commit to the wrong arm}) \leq 4 \exp \left(-\frac{m\Delta^2}{2} \right). \quad (4.7)$$

Taking, for example,

$$m \geq \frac{2 \log(4T)}{\Delta^2}$$

makes the wrong-commitment probability at most $1/T$. Ignoring constants,

$$m = O \left(\frac{\log T}{\Delta^2} \right).$$

This prescription is meaningful only when $2m \leq T$; otherwise there are no rounds left in which to exploit the final choice. The policy pays exploration regret while sampling arm 2, but makes the chance of a costly final mistake very small.

This is useful, but not satisfactory:

1. the exploration length depends on the unknown gap Δ ;
2. every arm receives a predetermined number of samples, even if it quickly becomes obviously poor;
3. the policy assumes a known horizon T ;
4. exploration is non-adaptive.

The ETC calculation exposes a sample-complexity principle that will recur:

Resolving a gap of size Δ costs about $1/\Delta^2$ samples.

The additional $\log T$ factor makes the conclusion hold with sufficiently high confidence over a horizon of length T .

Instead of fixing exploration in advance, can the data tell us *which* arm needs another sample and *when* enough has been learned?

4.4 Confidence intervals as guides

After n observations from arm i , Hoeffding gives the learner-visible interval

$$\left[\widehat{M}_{i,n} - r(n, \delta), \widehat{M}_{i,n} + r(n, \delta) \right]. \quad (4.8)$$

A narrow interval means the arm is well understood. A wide interval means that plausible values of its mean still vary substantially. This turns uncertainty into something the decision rule can use.

Suppose, for the moment, that each arm has a valid interval $[\widehat{M}_i - r_i, \widehat{M}_i + r_i]$. Which arm should we choose?

- **Greedy** chooses the largest empirical mean and ignores uncertainty.
- **Uniform or fixed exploration** keeps sampling according to a prearranged schedule, regardless of what the data show.
- **Optimism** chooses the arm whose true mean could plausibly be largest.

The optimistic index is the upper endpoint

$$\text{UCB}_i = \widehat{M}_i + r_i. \quad (4.9)$$

An arm has a high UCB because either it currently looks good—a reason to *exploit*—or it is still uncertain—a reason to *explore*. One number combines both motives.

Example 4.1 (A two-interval snapshot). Suppose the current confidence intervals are

$$[0.48, 0.62] \quad \text{and} \quad [0.35, 0.75].$$

The second arm has the lower empirical center but the larger upper endpoint. Optimism selects it because the available data have not yet ruled out that it is best. Another sample is valuable precisely because this uncertainty could change the decision.

4.4.1 What the confidence picture looks like

For a visual preview, take a known horizon T and use the radius

$$r_L(n) = \sqrt{\frac{2 \log T}{n}}.$$

The clean-event argument below derives why a logarithmic factor appears when one guarantee must cover many arms and sample sizes. The horizon-free UCB1 index instead uses the time-varying radius $r_A(n) = \sqrt{2 \log(n)/n}$ along a continuously sampled arm.

The left panel of Figure 4.1 centers an envelope at the unknown truth μ and asks where the random empirical-mean paths wander. The right panel centers intervals at the observed empirical mean and asks which values of μ remain plausible to the learner. Crossings are possible, but confidence bounds make them rare.

The interactive [Bandit Lab](#) lets you compare greedy, fixed exploration, and optimistic policies. Watch for the central feedback loop: sampling shrinks an interval, which changes which arm looks plausibly best, which changes what is sampled next.

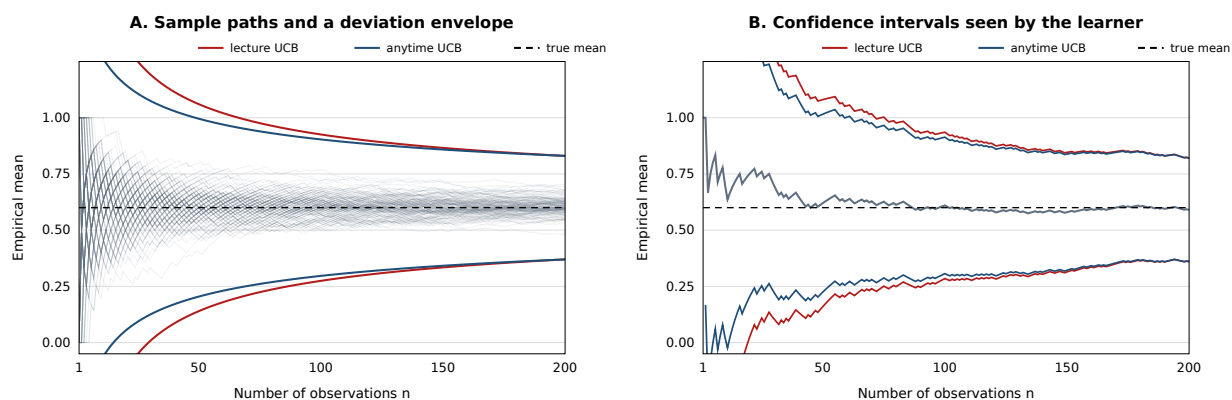


Figure 4.1. Two views of confidence for one Bernoulli arm with $\mu = 0.6$ and $T = 200$. The red *lecture UCB* radius is $r_L(n) = \sqrt{2\log(T)/n}$; the blue *anytime UCB* radius is $r_A(n) = \sqrt{2\log(n)/n}$ for $n \geq 2$. Left: 150 simulated empirical-mean paths and the two deviation envelopes around μ . Right: one realized path and the corresponding learner-visible intervals. The anytime radius is tighter before the horizon and meets the lecture radius at $n = T$. The figure uses a fixed random seed.

4.5 Simultaneous confidence and the clean event

Recall that there are K arms. Arm i has iid rewards in $[0, 1]$ with unknown mean μ_i . Imagine that arm i has an infinite reward tape

$$X_{i,1}, X_{i,2}, \dots, \quad \mathbb{E}[X_{i,s}] = \mu_i.$$

After observing the first n rewards from this arm, its empirical mean is

$$\widehat{M}_{i,n} = \frac{1}{n} \sum_{s=1}^n X_{i,s}. \quad (4.10)$$

The capital M emphasizes that $\widehat{M}_{i,n}$ is random before the data are observed; after observing a particular history, its realized value may be written $\widehat{m}_{i,n}$. The population mean μ_i is fixed but unknown.

Let

$$\mu^* = \max_i \mu_i, \quad \Delta_i = \mu^* - \mu_i.$$

Let

$$N_i(t) = \sum_{s=1}^t \mathbf{1}\{I_s = i\}$$

be the number of times arm i is selected in the first t rounds, and let $\widehat{M}_i(t) = \widehat{M}_{i,N_i(t)}$ be the empirical mean based on the rewards revealed from that arm by time t .

As in Lecture 3,

$$\text{Reg}(T) = \sum_i \Delta_i \mathbb{E}[N_i(T)].$$

Thus the central issue is to control how often suboptimal arms are played.

For a fixed arm and fixed sample size, (4.4) is enough. A bandit algorithm, however, repeatedly inspects estimates across K arms and all prefix lengths that might arise. Use the simple *lecture UCB* radius

$$r(n) = \sqrt{\frac{2 \log T}{n}}. \quad (4.11)$$

For each arm $i \in [K]$ and sample count $n \in \{1, \dots, T\}$, define the bad event

$$B_{i,n} = \left\{ |\widehat{M}_{i,n} - \mu_i| > r(n) \right\}. \quad (4.12)$$

There are exactly KT such events: one for each arm and each possible sample count. There is no additional index for calendar time. Hoeffding gives

$$\mathbb{P}(B_{i,n}) \leq 2 \exp\{-2nr(n)^2\} = 2T^{-4}.$$

The union bound therefore gives

$$\mathbb{P}\left(\bigcup_{i=1}^K \bigcup_{n=1}^T B_{i,n}\right) \leq 2KT^{-3}. \quad (4.13)$$

If $K \leq T$, this is at most $2/T^2$.

4.5.1 A clean event

Define

$$\mathcal{E} = \left\{ |\widehat{M}_{i,n} - \mu_i| \leq r(n) \text{ for every } i \in [K], n \in \{1, \dots, T\} \right\}. \quad (4.14)$$

Thus $\mathbb{P}(\mathcal{E}^c) \leq 2KT^{-3}$, and for $K \leq T$ the clean event has probability at least $1 - 2/T^2$. Conceptually, with overwhelmingly high probability, every empirical mean of every arm is accurate at every sample count that could arise.

At calendar time t , the empirical mean of arm i is

$$\widehat{M}_i(t) = \widehat{M}_{i,N_i(t)}.$$

The sample count $N_i(t)$ is random, but on $\mathcal{E}$ every possible prefix length $n = 1, \dots, T$ is already controlled. The clean event therefore covers whatever sample count the adaptive algorithm reaches; there is no need to union-bound separately over calendar time.

The payoff is that every subsequent step can be deterministic:

From this point on, conditional on $\mathcal{E}$, the proof is deterministic.

Probability proves that the clean event is overwhelmingly likely; optimization and regret reasoning are then carried out conditional on that event.

Remark 4.2 (Why the reward tape handles adaptivity). *The n th time arm i is selected, the algorithm reveals $X_{i,n}$ from that arm's tape. Adaptivity determines when the next reward is revealed; it does not change the distribution of the tape itself. We can therefore:*

1. *apply Hoeffding to every fixed prefix length n ;*
2. *union-bound over all prefix lengths and arms;*
3. *conclude that whichever random sample count $N_i(t)$ the algorithm reaches is already covered.*

This is why we do not apply a fixed-sample Hoeffding bound naively at the random sample size $N_i(t)$.

Remark 4.3 (Optional: anytime confidence *). *One elementary horizon-free construction allocates failure probability*

$$\delta_n = \frac{\delta}{Kn(n+1)}$$

to each arm at prefix length n . Since $\sum_{n=1}^{\infty} 1/[n(n+1)] = 1$, Hoeffding and a union bound give a confidence radius

$$r_{\text{any}}(n) = \sqrt{\frac{\log(2/\delta_n)}{2n}} = O\left(\sqrt{\frac{\log(Kn/\delta)}{n}}\right)$$

that is valid for every arm and every $n \geq 1$ simultaneously.

The tighter blue radius plotted in Figure 4.1 is the time-varying index used by the UCB1 algorithm:

$$r_i^A(t) = \sqrt{\frac{2 \log t}{N_i(t)}}. \quad (4.15)$$

It also gives logarithmic regret without knowing T , but its proof is more delicate than conditioning on our one fixed-horizon clean event: it controls bad estimation events round by round and sums their probabilities. This is the classical analysis of Auer, Cesa-Bianchi, and Fischer (2002).

4.6 The formal UCB algorithm

On the clean event, all we know about arm i is that

$$\mu_i \in \left[\widehat{M}_i(t-1) - r(N_i(t-1)), \widehat{M}_i(t-1) + r(N_i(t-1)) \right].$$

We now make the optimistic rule developed above precise. Define

$$\text{UCB}_i(t) = \widehat{M}_i(t-1) + r(N_i(t-1)), \quad (4.16)$$

$$\text{LCB}_i(t) = \widehat{M}_i(t-1) - r(N_i(t-1)). \quad (4.17)$$

The UCB algorithm is:

1. Play each arm once.
2. For $t > K$, choose

$$I_t \in \arg \max_i \text{UCB}_i(t). \quad (4.18)$$

Why is this sensible? An arm has a large UCB for one of two reasons:

- its empirical mean is large — a reason to *exploit*;
- its confidence radius is large — a reason to *explore*.

UCB combines these two motives in a single index.

4.7 The key UCB inequality

Let i^* be an optimal arm. Suppose UCB chooses arm i at round t . By definition,

$$\text{UCB}_i(t) \geq \text{UCB}_{i^*}(t).$$

On the clean event,

$$\text{UCB}_{i^*}(t) \geq \mu^*.$$

Also,

$$\widehat{M}_i(t-1) \leq \mu_i + r(N_i(t-1)),$$

so

$$\text{UCB}_i(t) = \widehat{M}_i(t-1) + r(N_i(t-1)) \leq \mu_i + 2r(N_i(t-1)).$$

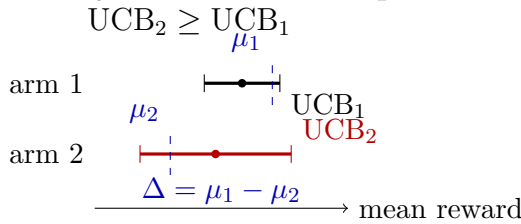
Therefore

$$\mu^* \leq \mu_i + 2r(N_i(t-1)), \quad (4.19)$$

or

$$\boxed{\Delta_i \leq 2r(N_i(t-1))}. \quad (4.20)$$

A. Early: intervals overlap



B. Later: arm 2 is resolved

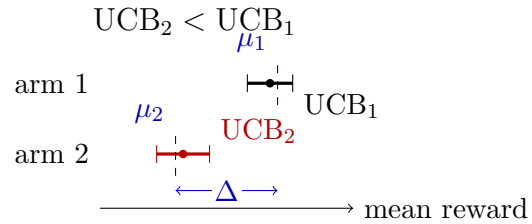


Figure 4.2. Why UCB stops exploring a suboptimal arm. Solid segments are learner-visible confidence intervals, dots are empirical means, and dashed blue markers are the true means. While the intervals overlap, optimism may select arm 2. Once its radius is small relative to the true gap Δ , its UCB falls below that of the optimal arm. On the clean event, selecting arm 2 requires $\Delta \leq 2r_2$.

This is the whole proof in one line:

If UCB still chooses a bad arm, then that arm must still be sufficiently uncertain.

Substituting (4.11) into (4.20),

$$\Delta_i \leq 2\sqrt{\frac{2 \log T}{N_i(t-1)}}.$$

Rearranging,

$$N_i(t-1) \leq \frac{8 \log T}{\Delta_i^2}. \quad (4.21)$$

Thus, apart from initialization and the rare bad event, a suboptimal arm cannot be selected after it has accumulated more than order $\log T / \Delta_i^2$ samples. Its final count can exceed the displayed threshold by at most the last selected round.

4.8 Gap-dependent regret

Using the regret decomposition,

$$\text{Reg}(T) = \sum_{i:\Delta_i>0} \Delta_i \mathbb{E}[N_i(T)].$$

Equation (4.21) suggests

$$\mathbb{E}[N_i(T)] = O\left(1 + \frac{\log T}{\Delta_i^2}\right).$$

Hence

$$\boxed{\text{Reg}(T) = O\left(\sum_{i:\Delta_i>0} \frac{\log T}{\Delta_i}\right)}. \quad (4.22)$$

The contribution of the bad event is negligible because regret is at most T while $\mathbb{P}(\mathcal{E}^c) = O(T^{-2})$.

Remark 4.4 (Instance-dependent versus worst-case guarantees). *The bound (4.22) is excellent when every suboptimal arm is separated from the optimum by a nontrivial gap. But it becomes large when some Δ_i is very small. A complementary argument splits arms into “small-gap” and “large-gap” groups and yields an instance-independent bound of order*

$$O\left(\sqrt{KT \log T}\right).$$

Sharp constants are secondary here; the important point is the qualitative difference between logarithmic instance-dependent regret and square-root worst-case regret.

4.9 Optional: successive elimination *

UCB is not the only way to use confidence intervals. Another natural algorithm repeatedly samples all active arms and removes arm i once

$$\text{UCB}_i(t) < \text{LCB}_j(t)$$

for some other active arm j .

The same clean-event proof implies that the best arm is never eliminated, and an arm with gap Δ_i is discarded after roughly

$$O\left(\frac{\log T}{\Delta_i^2}\right)$$

samples.

This alternate view is useful because it isolates the deeper principle:

adaptive exploration means spending samples only until an action is statistically resolved.

4.10 Optional: why exploration is unavoidable *

Could a clever algorithm avoid trying uncertain arms altogether? No. To know whether an arm with mean $1/2$ is actually slightly better or slightly worse than another arm, one needs enough samples to distinguish nearby probability distributions.

For a Bernoulli coin with mean

$$\frac{1}{2} + \varepsilon$$

versus one with mean

$$\frac{1}{2} - \varepsilon,$$

the standard error after n samples is of order $1/\sqrt{n}$. Distinguishing the two therefore requires roughly

$$n = \Omega\left(\frac{1}{\varepsilon^2}\right)$$

samples. More refined lower bounds use KL divergence, but the core message is already visible here:

learning a gap of size ε costs on the order of $1/\varepsilon^2$ observations.

This explains why the UCB sample complexity in (4.21) has exactly that dependence, up to the logarithmic confidence factor.

4.11 Back to pricing: prices as arms

Suppose a seller chooses from a finite price set

$$\mathcal{P} = \{p_1, \dots, p_K\}.$$

At round t , price p_i is offered. Let

$$Y_t = \mathbf{1}\{\text{customer purchases at round } t\}.$$

Conditional on choosing price p_i ,

$$Y_t \sim \text{Bernoulli}(q_i), \quad q_i = 1 - F(p_i),$$

where q_i is unknown to the seller. The realized revenue and its conditional mean are

$$X_t = p_i Y_t, \quad \mu_i = p_i q_i.$$

After normalizing by the largest possible price, this is exactly a stochastic bandit.

UCB is a canonical model for the exploration–exploitation phenomenon. It should not be read as a claim that a production pricing system literally treats every price as an unrelated arm; the structured feedback discussed below often supports more specialized algorithms.

So a direct pricing implementation of UCB is simply:

$$\boxed{\text{offer the price with the largest optimistic estimate of expected revenue.}} \quad (4.23)$$

This captures the idea that

$$\boxed{\text{a price is both a revenue decision and an experiment.}}$$

4.12 But pricing has extra structure

Treating each candidate price as an unrelated arm throws away useful information.

Suppose a customer has a fixed valuation V . If we offer price p and observe a sale, then we learn

$$V \geq p.$$

This implies that the customer would also have purchased at every lower price. Conversely, if there is no sale at p , then

$$V < p,$$

so the customer would not have purchased at any higher price.

These are counterfactual statements about the same customer. Under an iid valuation model, they are also censored information about F , and hence about purchase probabilities at other prices. Thus pricing has *partial feedback* across prices:

- sale at p gives information about all lower prices;
- no sale at p gives information about all higher prices.

There is also functional structure. The mean revenue is not an arbitrary vector:

$$\mu(p) = p(1 - F(p)).$$

Nearby prices typically have related purchase probabilities and revenues.

The optional binary-logit model in Lecture 3 imposes still more structure:

$$q(p, x) = \sigma(\theta^\top x - \beta p).$$

One observation can then update common parameters and inform many prices at once. This suggests a useful hierarchy:

$$\boxed{\text{unstructured finite arms} \longrightarrow \text{parametric demand} \longrightarrow \text{ordered or continuum prices.}}$$

Moving right can improve learning by sharing information, but only when the additional modeling assumptions are credible.

The basic UCB reduction ignores both forms of structure. More specialized dynamic-pricing algorithms exploit them.

4.13 Optional: a glimpse of continuum pricing ^{*}

Suppose prices can be any value in an interval, say $p \in [0, 1]$. One simple idea is to discretize into K grid points and run a finite-armed bandit algorithm.

This creates two sources of regret:

1. **learning error:** regret from not knowing which grid price is best;
2. **discretization error:** regret because the true optimal price may lie between grid points.

If the revenue curve is sufficiently smooth, a finer grid reduces discretization error but creates more arms to learn. Choosing the grid therefore involves a bias–variance-like tradeoff.

This is the starting point of the dynamic-pricing literature exemplified by Kleinberg–Leighton: exploit the order and smoothness of prices rather than treating prices as unrelated actions. We will not develop the full theory here; the finite-armed model already gives us the core exploration–exploitation ideas that recur later in the course.

4.14 Proof pattern to remember

The UCB proof illustrates a template we will use again:

1. define an ideal benchmark;
2. construct an empirical estimate and a high-probability “clean event”;
3. show that a wrong decision on the clean event can happen only while some uncertainty remains large;
4. bound the number of such decisions;
5. charge the rare complement of the clean event separately.

More rounds shrink uncertainty only when the decision rule continues to generate informative observations. UCB is one way to convert scale into learning: uncertainty itself forces the samples needed to resolve it.

Later, the Bayes Selector analysis will use a different algorithm and benchmark, but a related high-level idea: control regret by controlling the probability or frequency with which the online action disagrees with an ideal offline decision.

4.15 Takeaways

1. The CLT suggests that empirical-mean uncertainty shrinks like $1/\sqrt{n}$.
2. Hoeffding turns that intuition into a finite-sample, high-probability guarantee.
3. The union bound combines failure probabilities without requiring independence.
4. Explore then commit shows why resolving a gap Δ costs about $1/\Delta^2$ samples, plus a logarithmic factor for high confidence.
5. Fixed exploration is wasteful because it cannot react to what has already been learned.
6. Confidence intervals suggest optimism: sample the arm whose mean could plausibly be largest.
7. A clean event makes all the confidence statements needed by the adaptive algorithm valid simultaneously.
8. UCB uses uncertainty itself to trigger exploration.
9. On the clean event, choosing a suboptimal arm implies $\Delta_i \leq 2r_i$.
10. This yields logarithmic instance-dependent regret.
11. Pricing has additional order, partial-feedback, and parametric structure beyond a generic bandit.

Recommended reading

Core / closest to these lectures.

Slivkins, Chapter 1, Sections 1.3.1–1.3.3. These sections develop the reward-tape clean event, confidence bounds, successive elimination, and optimism/UCB in essentially the proof style used here. The CLT-to-Hoeffding and explore-then-commit discussion and the sample-path figure are course-note exposition connecting earlier probability intuition to that proof.

Alternative perspective.

Re-read Lecture 3’s pricing formulation while translating every arm i into the action “charge price p_i .” This makes clear why the generic UCB theorem applies after revenue normalization, and what economic information the abstract arm model suppresses.

Optional / deeper.

Slivkins, Chapter 2, gives lower bounds showing that exploration is unavoidable. Auer, Cesa-Bianchi, and Fischer, “[Finite-time Analysis of the Multiarmed Bandit Problem](#)” (2002), gives the classical UCB1 algorithm with its time-varying radius.